

1 Image pre-processing

Gaussian
and median
filter

intensity
normalization
across z-depth

Richardson-Lucy
deconvolution

2 DisPerSe skeleton

compute the
1-manifold

simplify by
persistent
homology

3 Skeleton processing

redefine
junctions

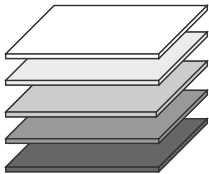
join
filaments

join
junctions

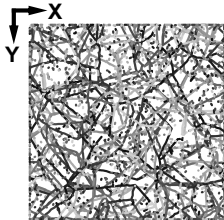
clean
dangling
filaments

convert
to network
graph

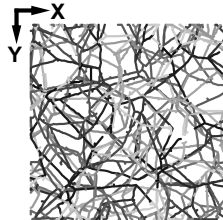
z



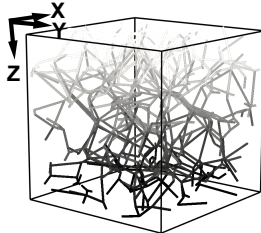
processed image stack



fiber network skeleton



processed
fiber network skeleton



characterization of network
morphology and topology